Day 9 Activity: AI-assisted analysis, with verification

Friday, May 29, 2026

Your Name Here

AI Disclosure

You are **required** to use AI for at least part of this activity. List every AI tool you used and roughly how you used it:

- ☐ ChatGPT (any version)
- ☐ Claude (any version)
- ☐ Gemini (any version)
- ☐ GitHub Copilot
- ☐ Other: ______________________

Briefly describe your use (2-3 sentences). Be specific about what you asked.

[Type your explanation here.]

Setup

```r
library(tidyverse)
library(nycflights13)
```

We're using the `nycflights13` package, a dataset of every commercial flight that departed from one of NYC's three major airports in 2013. You've already seen `flights` on Homework 2. Today we'll also be using a second table, `airlines`, which gives the full names of each carrier.

```r
glimpse(flights)
```

```
Rows: 336,776
Columns: 19
$ year           <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
$ month          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ day            <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ dep_time       <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
$ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
$ dep_delay      <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -
1~
$ arr_time       <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,~
$ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,~
$ arr_delay      <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -
1~
$ carrier        <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
$ flight         <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
$ tailnum        <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
$ origin         <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA",~
$ dest           <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD",~
$ air_time       <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
$ distance       <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
$ hour           <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
$ minute         <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
$ time_hour      <dttm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-
01 0~
```

```r
glimpse(airlines)
```

```
Rows: 16
Columns: 2
$ carrier <chr> "9E", "AA", "AS", "B6", "DL", "EV", "F9", "FL", "HA", "MQ", "O~
$ name    <chr> "Endeavor Air Inc.", "American Airlines Inc.", "Alaska Airline~
```

The task

You're working with a regional newspaper. A reporter asks you:

> **"Which airlines were the worst for travelers leaving JFK in summer 2013? I want a clear, readable graphic and a couple of sentences I can quote."**

Their definition of "worst" is intentionally vague. That's part of the assignment; different reasonable interpretations could give different answers. Your job has three parts.

Part 1

Use any AI tool you want. Ask it for help building an analysis that answers the reporter's question. You should end up with:

- A pipeline that filters to JFK summer 2013 flights
- Some measure of "worst"
- The airline name (not the 2-letter code)
- A finished plot with proper labels
- 2-3 sentences interpreting it

Paste the AI's response below as-is. Do not edit it yet. You'll edit in Part 3.

```
# Paste AI's code here. Try to run it as-is, even if you suspect it won't work.
```

What did the AI claim about the answer? Copy or paraphrase its interpretation:

[Paste here.]

Part 2

Now work through the AI's output critically. Find at least **three** things that are wrong, suspicious, unverified, or could go wrong. For each one, write:

1. **What's the problem?**
2. **How did you find it?** (Did you read the code? Run a check? Compare to the data?)
3. **What's the right answer or fix?**

Things to check

- Did it filter "summer" correctly? What months count? Did it filter on `month` or did it try something weird with dates?
- Did it handle `NA` values? `flights$dep_delay` has a lot of NAs (cancelled flights). Are they being dropped silently, or treated as 0, or treated as NA in the answer?
- Did it join `flights` to `airlines` to get the carrier name? Did the join work as expected?
- Did it average delays, sum them, or count delayed flights? Are those the same question?
- Did the AI invent any functions or arguments that don't exist? Run `?function_name` on anything you haven't seen.
- Does the plot have axis labels? Units? A title that reflects what's actually being shown?
- Did the AI claim something in its interpretation that isn't actually visible in the plot?

Issue 1

Problem:

[Describe.]

How you found it:

[Describe.]

Fix:

[Describe.]

Issue 2

Problem:

[Describe.]

How you found it:

[Describe.]

Fix:

[Describe.]

Issue 3

Problem:

[Describe.]

How you found it:

[Describe.]

Fix:

[Describe.]

Issue 4 (optional)

[If you found more, list them here.]

Part 3

Now produce **your own** version of the analysis, with the issues fixed. Use the AI's code as a starting point if you want, but **every line should be one you understand and can defend**.

```
# Your verified pipeline:
```

```
# Your verified plot:
```

Your interpretation (2-3 sentences):

[Write here. State explicitly what definition of "worst" you used and any caveats the reporter should know.]

Part 4

In **3-4 sentences**, answer the following:

1. What was the most surprising thing the AI got right or wrong?
2. If you had submitted the AI's first version without review, what would have been the worst-case consequence?
3. How would you change the way you talk to AI in the future based on what you found?

[Your reflection here.]

Code appendix

```
library(tidyverse)
library(nycflights13)
glimpse(flights)
glimpse(airlines)
# Paste AI's code here. Try to run it as-is, even if you suspect it won't work.

# Your verified pipeline:

# Your verified plot:
```